

JavaScript Advanced Questions

JavaScript Fundamentals

1. What is JavaScript, and how is it different from Java?
2. What are the different data types in JavaScript?
3. What is the difference between null and undefined?
4. What is the difference between == and ===?
5. What are truthy and falsy values in JavaScript?
6. What is type coercion?
7. What is NaN? How can you check if a value is NaN?
8. What is the difference between let, const, and var?
9. What is variable hoisting?
10. What is the temporal dead zone (TDZ)?

Functions

11. What is a function declaration vs. a function expression?
12. What are arrow functions, and how do they differ from regular functions?
13. What is the this keyword in JavaScript?
14. How does this behave inside an arrow function?
15. What is a callback function?
16. What are higher-order functions?
17. What is a closure? Explain with an example.
18. What is currying in JavaScript?
19. What is function borrowing?
20. What are IIFEs (Immediately Invoked Function Expressions)?

Scope & Execution Context

21. What is lexical scope?
22. What is the scope chain?
23. What is the execution context?
24. What is the call stack?
25. What is the difference between global scope, function scope, and block scope?

Objects & Arrays

26. What is the difference between primitive and reference data types?
27. How do you create an object in JavaScript?
28. What is object destructuring?
29. What is the spread operator (...)?
30. What is the rest parameter?
31. What is the difference between map(), filter(), and reduce()?
32. What is the difference between forEach() and map()?
33. How do shallow copy and deep copy differ?
34. What are object methods?
35. How can you iterate through object properties?

Asynchronous JavaScript

36. What is synchronous vs. asynchronous programming?
37. What is the event loop?
38. What are Promises?
39. What are the states of a Promise?
40. What is the difference between callbacks and Promises?
41. What is async/await?
42. How does Promise.all() work?
43. What is the difference between Promise.all() and Promise.allSettled()?
44. What are microtasks and macrotasks?
45. Explain the execution order of:

```
console.log("1");
```

```
setTimeout(() => console.log("2"), 0);
```

```
Promise.resolve().then(() => console.log("3"));
```

```
console.log("4");
```

Advanced JavaScript

46. What is prototypal inheritance?
47. What is the prototype chain?
48. What are ES6 modules? Explain import and export.

49. What is event delegation?

50. What is the difference between debounce and throttle?